

MULTI-TRAIT HUMAN PROFILING FROM FINGERPRINTS USING DEEP LEARNING MODELS

A FINAL YEAR PROJECT REPORT

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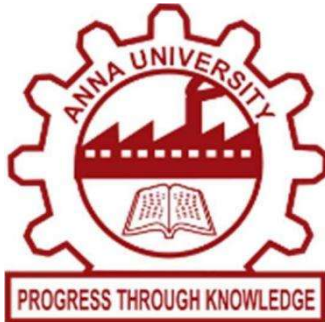
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BONAFIDE CERTIFICATE

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who carried out the project in the Department of Electronics Engineering, Madras Institute of Technology, Anna University Chennai 600 044. The work reported here is original and does not form part of any other project report on the basis of which a degree or award was conferred on an earlier occasion. We understand the University’s policy on plagiarism and declare that the project report is our own work, except where specifically acknowledged and has not been copied from other sources or been previously submitted.

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ABSTRACT

Traditional methods for determining biological traits such as blood group, finger type and gender are invasive, time-consuming, and infeasible in emergency scenarios. This research proposes a novel, non-invasive framework for predicting blood group, gender, and finger type simultaneously through deep learning-based fingerprint analysis.

Fingerprints serve as a unique and permanent biometric marker, providing an ideal foundation for extracting critical personal information without the need for physical samples. The core of this system is two dedicated 22-layer Convolutional Neural Network (CNN) models. The first model was designed for an 8-class blood group classification (A+, A-, B+, B-, AB+, AB-, O+, O-) and achieved a train accuracy of 97.8% on a dataset of 8,000 images. The second 22-layer CNN was developed for the simultaneous prediction of gender and specific finger type (20 classes) using 55,000 images, achieved train accuracy of 96.74%, demonstrating robust generalization with validation accuracy exceeding training accuracy. A critical finding was that our optimized custom CNNs outperformed transfer learning with complex models like InceptionV3 and ResNet-50, which are suboptimal for fingerprint-specific feature extraction, by achieving an optimal balance of high accuracy and computational efficiency.

The trained models were integrated and deployed via a user friendly Streamlit web application, providing a real-time prediction interface. This system eliminates the need for physical blood samples and offers a rapid, reliable alternative for preliminary screening. The results confirm a significant correlation between fingerprint patterns and biological traits, paving the way for transformative applications in forensic science, emergency healthcare, and advanced biometric security systems.

Keywords: Deep Learning, Convolutional Neural Network (CNN), Fingerprint Analysis, Blood Group Prediction, Gender Classification, Biometrics, Non-Invasive Screening.

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LIST OF ABBREVIATION

3D	- 3 Dimension
AI	- Artificial Intelligence
AFIS	- Automated Fingerprint Identification System
CNN	- Convolutional Neural Network
DL	- Deep Learning
F1-score	- Harmonic Mean of Precision and Recall
GeLU	- Gaussian Error Linear Unit
GLCM	- Gray-Level Co-occurrence Matrix
GPU	- Graphics Processing Unit
GUI	- Graphical User Interface
HOG	- Histogram of Oriented Gradients
IAFIS	- Integrated Automated Fingerprint Identification System
KNN	- K-Nearest Neighbors
LBP	- Local Binary Patterns
MDSVM	- Multi-Dimensional Support Vector Machine
ML	- Machine Learning
RBC	- Red Blood Cell
RBF	- Radial Basis Function
ReLU	- Rectified Linear Unit
RF	- Random Forest
RVR	- Ridge-to-Valley Ratio
SGD	- Stochastic Gradient Descent
SVM	- Support Vector Machine
VGG16	- Visual Geometry Group of the University of Oxford

CHAPTER 1

INTRODUCTION

Fingerprints have always been one of the most reliable and distinctive forms of human identification. Each individual's fingerprint pattern is unique and remains unchanged throughout their lifetime, making it a vital tool in forensic science, law enforcement, and personal authentication. In recent years, the scope of fingerprint analysis has expanded beyond identity verification to include the prediction of human traits such as gender, blood group, and other physiological characteristics. This project, "Multi-Trait Human Profiling from Fingerprints Using Deep Learning Models" explores the potential of using modern deep learning techniques to extract meaningful features from fingerprint images and relate them to multiple human traits. By combining advanced image processing, feature extraction, and deep neural architectures like Convolutional Neural Networks (CNNs), the system aims to achieve accurate, non-invasive, and automated profiling based solely on fingerprint patterns.

1.1 MOTIVATION

Traditional fingerprint analysis has focused mainly on identification and verification through manual observation or pattern matching. With advancements in artificial intelligence, deeper correlations within fingerprint patterns can now be explored. Studies have shown links between fingerprints and traits like gender and blood group, but most approaches handle them separately. This project aims to develop a unified deep learning model that can predict multiple human traits from a single fingerprint, offering a non-invasive and efficient solution for forensic, medical, and biometric applications.

1.2 OBJECTIVE

The main objectives of this project are as follows:

- ❖ To develop a deep learning model for predicting multiple human traits from fingerprints.
- ❖ To analyze the relationship between fingerprint patterns like gender, finger-type and blood group.
- ❖ To create a non-invasive and automated system for human profiling.

1.3 TRADITIONAL METHODS

1.3.1 BLOOD GROUP IDENTIFICATION

Blood group identification is historically performed using serological techniques, where blood samples are mixed with reagent antibodies to observe agglutination reactions. Methods such as the tube technique, gel column method, and slide testing are standard laboratory procedures for determining the ABO and Rh blood groups by detecting antigens and antibodies on the surface of red blood cells. These approaches form the backbone of clinical transfusion medicine and have been refined to maximize accuracy and reliability, although they require biological samples and careful manual implementation.

A. TUBE TECHNIQUE

The tube method is a conventional serological technique used to determine blood groups by mixing a suspension of Red Blood Cells (RBCs) with specific antisera in a test tube and observing agglutination reactions shown in Fig 1.1 (a). The process involves forward typing, where the RBCs are tested against anti-A, anti-B, and anti-AB antibodies, and reverse typing, where serum is tested against known RBC antigens. After gently mixing the reagents with the cells, the mixture is centrifuged to facilitate antigen-antibody interaction. Agglutination, visible as clumping or sedimentation of RBCs, indicates the presence of corresponding antigens. This method is more sensitive and reliable than slide testing and is widely used for initial and emergency blood typing, although it requires careful technique, including proper centrifugation and interpretation under ambient conditions.

B. GEL-COLUMN METHOD

The gel column method is a modern advancement in blood group identification that uses microcolumns filled with a gel matrix designed to trap agglutinated red blood cells during centrifugation. Blood samples, usually plasma or serum mixed with red blood cells (RBCs), are added to these gel-filled microtubes or cards. When antigen-antibody reactions occur, the clumped RBCs are trapped at the top or within the gel, while non-agglutinated cells pass through to the bottom. This separation by size allows for clear visual interpretation of results, as the

location of red cells after centrifugation directly indicates the presence or absence of agglutination, marking positive or negative blood group reactions. This technique is highly sensitive, reproducible, and easier to standardize compared to traditional tube methods. It reduces false positives and negatives typical of manual tube testing and minimizes human error related to subjective interpretation. The gel method shown in Fig 1.1 (b), is widely accepted in clinical laboratories and blood banks and is often integrated with automated blood grouping systems for high-throughput, fast, and reliable analysis.

C. SLIDE TESTING

Slide testing is a simple and quick method suitable for point-of-care or emergency blood group screening. A drop of the patient's blood (usually a finger prick) is placed on a glass slide as shown in Fig 1.1 (c), then mixed with a drop of specific antisera (anti-A, anti-B, or anti-D). The mixture is gently rocked or stirred, and within seconds to a minute, the presence or absence of agglutination is observed visually by the naked eye. Agglutination appears as visible clumping or granules, whereas a uniform suspension indicates no reaction. Despite its speed and simplicity, slide testing is less accurate and sensitive compared to tube or gel methods. It is prone to errors caused by inadequate mixing, environmental factors (temperature, humidity), and subjective visual interpretation. Weak reactions or subgroups may be missed, and improper reagent volumes can cause false results. Therefore, it is recommended only for initial screening or in situations where rapid preliminary results are needed, with confirmatory testing performed afterward.

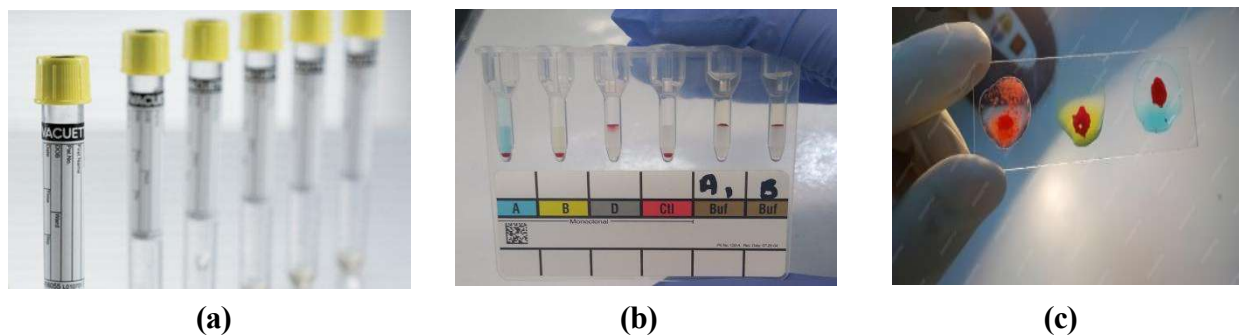


Fig 1.1 Blood group identification methods

a) Tube technique, b) Gel column method, c) Slide testing.

1.3.2 GENDER IDENTIFICATION

Gender identification from fingerprints has its roots in the field of dermatoglyphics, where experts visually analyze the physical characteristics of fingerprint patterns. Traditional methods rely on measuring ridge density, pattern type prevalence, and morphological features, such as whorl or loop frequency, which can show statistical differences between male and female populations. While not definitive on an individual basis, these methods are widely used in forensic anthropology and population studies for preliminary gender assessment.

A. RIDGE DENSITY

Ridge density refers to the count of the ridges within a defined area of the fingerprint, typically measured as the number of ridges per 25 square millimeters. Studies have shown that females generally have a higher ridge density than males, which means more ridges are packed into a smaller area on the finger. The counting involves manually selecting a region on the fingerprint and counting the number of ridges crossing the area to generate a quantitative value. This parameter is often used as a biometric marker to suggest probable gender. The different types of ridge density is shown in Fig 1.2 (a).

B. PATTERN TYPE

The pattern type focuses on the global fingerprint pattern whether it is a loop, whorl, or arch as shown in Fig 1.2 (b). Research indicates variation in the prevalence of these patterns between genders; for example, whorls and arches may be more frequent in males while loops might occur more in females depending on the studied population. These pattern frequencies serve as statistical indicators in gender classification, although overlap exists on an individual basis. By categorizing fingerprints into these major groups and analyzing their distribution, forensic experts can contribute to gender estimation.

C. MORPHOLOGICAL FEATURES

Morphological features include other finer aspects of the fingerprint such as ridge thickness, ridge-to-valley ratio, minutiae distribution, and ridge pattern complexity. For instance,

males tend to have thicker and more widely spaced ridges compared to females. Morphological observations also extend to ridge shape, pore placement, and asymmetry. These features shown in Fig 1.2 (c), require detailed analysis under magnification and can supplement ridge density and pattern type findings to improve gender identification reliability.

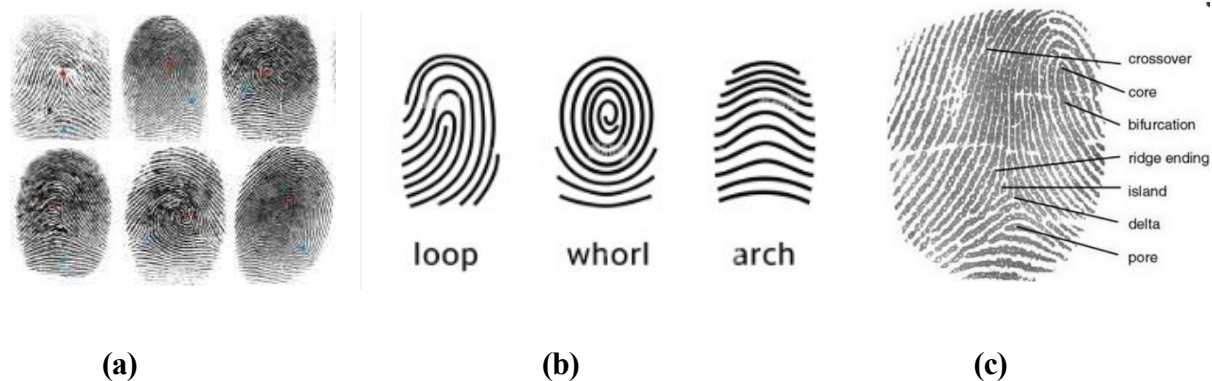


Fig 1.2 Gender and Finger-type identification methods
(a) Ridge density, (b) Pattern types, (c) Morphological features.

1.3.3 FINGERTYPE CLASSIFICATION

Traditional fingertype classification is performed using manual systems like the Henry Classification System, which categorizes fingerprints into arches, loops, and whorls based on their global pattern and core features. This system involves expert inspection of the ridge flow, identification of key reference points such as deltas and cores, and detailed comparison with established standards. It has been the cornerstone of personal identification and forensic science for over a century and continues to be employed for organizing and matching fingerprint records.

The classification involves expert inspection of ridge patterns to identify key reference points such as cores and deltas, and detailed comparison with established standards. The Henry system reduces the complexity of fingerprint record management by grouping prints into primary categories and refining them further with secondary, sub-secondary, major, final, and key classifications. This layered approach has made it the cornerstone of personal identification and forensic science for over a century. Despite the emergence of automated systems, it continues to be widely employed globally for organizing and matching fingerprint records efficiently through

a structured framework. This classification method enables exclusion of non-matching prints quickly by sorting fingerprint sets into coarse pattern-based categories before microscopic feature comparison, significantly speeding up the identification process in law enforcement and forensic investigations.

1.4 MODERN FINGERPRINT IDENTIFICATION METHODS

Modern fingerprint identification has transformed traditional manual methods through digital technology, advanced imaging, and computational algorithms. These systems enable fast, accurate recognition for law enforcement, border security, civil ID, and access control. Unlike manual classification based on general patterns, modern methods extract detailed local features (minutiae) and use automated algorithms to match fingerprints against large databases within seconds. High-resolution sensors optical, capacitive, and ultrasonic capture fingerprints, while image enhancement ensures reliable feature detection. Extracted minutiae are encoded into templates for efficient comparison in systems like AFIS and IAFIS, which provide real-time, nationwide identification. Emerging technologies like 3D and contactless sensing further improve accuracy, usability, and hygiene, making modern fingerprint identification a key pillar of security and forensic systems worldwide.

A. AUTOMATED FINGERPRINT IDENTIFICATION SYSTEMS (AFIS)

Automated Fingerprint Identification System (AFIS) is a sophisticated biometric technology that digitizes, stores, and compares fingerprint data for rapid identification. Originally developed to automate the labor-intensive manual fingerprint matching process, AFIS extracts unique fingerprint features called minutiae points ridge endings, bifurcations, and other distinctive characteristics from digital fingerprint images. AFIS can scan millions of fingerprint records in seconds and generate a shortlist of possible matches, significantly accelerating criminal investigations, background checks, and identity verification processes. The system integrates hardware for fingerprint capture, image processing software, large biometric databases, and matching algorithms as depicted in Fig 1.3 (a). High-resolution scanners capture rolled or plain impressions, while image processing algorithms enhance clarity and isolate minutiae features. The extracted data is compressed into mathematical templates enabling fast

comparison against stored records. AFIS has diverse applications beyond law enforcement, including civil identification, border control, and financial security systems. Continuous enhancements incorporating machine learning improve accuracy, especially with partial or low-quality fingerprints, making AFIS a cornerstone of modern biometric identification.

B. IMAGE ACQUISITION AND SENSOR TECHNOLOGIES

Accurate fingerprint identification begins with high-quality image acquisition, enabled by advanced sensor technologies. Common acquisition methods include optical scanners shown in Fig 1.3 (b), which use light reflection; capacitive sensors, which detect variations in electrical charge; ultrasonic sensors, which use sound waves to capture three-dimensional ridge details; and thermal sensors, which detect temperature differences. The choice of sensor affects image resolution, durability, and susceptibility to external conditions such as dirt or moisture.

Modern fingerprint scanners vary from flat platen devices used for plain impressions to live-scan rolling devices capturing full fingerprint details. Ultrasonic and capacitive sensors provide enhanced imaging accuracy by capturing subdermal features and reducing spoofing risks. 3D and multispectral imaging technologies have further advanced acquisition by penetrating skin layers to capture detailed ridge structures beneath the surface. These technologies address challenges of poor-quality fingerprints and distortions, facilitating more reliable downstream processing in fingerprint recognition systems.

C. FINGERPRINT IMAGE ENHANCEMENT AND PRE-PROCESSING

Fingerprint image enhancement improves the quality of acquired images to facilitate reliable feature extraction and matching. Pre-processing techniques include noise reduction, contrast adjustment, and image normalization to compensate for poor lighting or sensor inconsistencies. Segmentation isolates the fingerprint region from the background, and binarization converts grayscale images into binary form to delineate ridges and valleys sharply. Advanced methods such as Gabor filtering and histogram equalization emphasize ridge-valley structures and reduce noise artifacts, resulting in clearer ridge patterns. Image thinning algorithms reduce ridge width to a single pixel line, enabling more precise detection of minutiae points. Effective enhancement and pre-processing are crucial for accurate fingerprint analysis,

especially when dealing with latent prints or low-resolution images obtained from non-ideal conditions.

D. MINUTIAE EXTRACTION AND FEATURE DETECTION

Minutiae extraction is the process of detecting unique fingerprint ridge characteristics such as ridge endings and bifurcations, which serve as primary biometric features for identification. Algorithms analyze thinned and enhanced fingerprint images to locate these points based on local ridge directions and geometric properties. The extracted minutiae points form the fingerprint template used for comparison in identification systems.

In addition to minutiae, feature detection can include pore extraction and ridge flow orientation, providing richer biometric information. Robust minutiae extraction algorithms must handle fingerprint distortions, partial prints, and noise to avoid false minutiae that can reduce matching accuracy. Advanced machine learning approaches also assist in differentiating true minutiae from artifacts, improving the reliability and robustness of fingerprint recognition systems.

E. PATTERN RECOGNITION AND MATCHING ALGORITHMS

Pattern recognition algorithms categorize fingerprints based on global ridge patterns such as loops, whorls, and arches, assisting in coarse classification to reduce search space during matching as shown in Fig 1.3 (c). Matching algorithms compare the extracted minutiae templates of query and database fingerprints using metrics like spatial relationships, distance, and geometric transformations. Sophisticated matching algorithms incorporate tolerance to translation, rotation, and deformation, addressing real-world variations in fingerprint presentation. Techniques include ridge alignment, minutiae-based matching, correlation-based methods, and graph matching. Recent advances utilize machine learning and deep neural networks to improve matching performance under challenging conditions, such as partial or low-quality prints, significantly enhancing accuracy and speed in fingerprint identification.

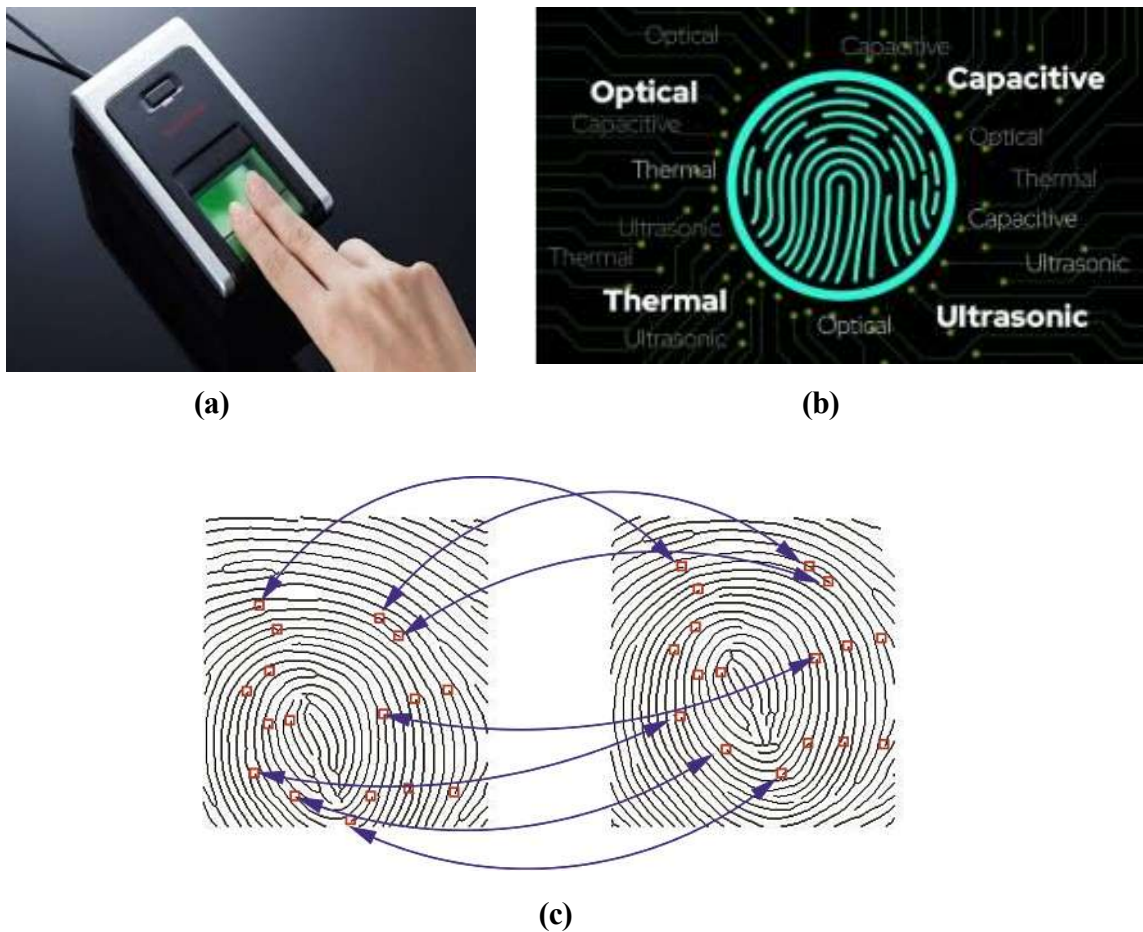


Fig 1.3 Modern fingerprint methods.

(a) AFIS, (b) Sensor technologies, (c) Pattern recognition.

F. INTEGRATED AUTOMATED FINGERPRINT IDENTIFICATION SYSTEM (IAFIS)

The Integrated Automated Fingerprint Identification System (IAFIS) is a large-scale, nationwide biometric network used across the United States, designed to support rapid and accurate fingerprint-based identification. Operated by the FBI, it links together fingerprint databases from numerous federal, state, and local agencies into one unified system. This interconnected framework allows for seamless data sharing and ensures that fingerprint records can be searched, matched, and verified within seconds, regardless of where they were originally collected.

IAFIS maintains millions of fingerprint templates along with associated demographic information, making it one of the most comprehensive biometric repositories in the world. Its ability to perform fast comparisons and return reliable results plays a crucial role in a wide range of applications, including criminal investigations, civil background checks, immigration processing, and employment screening. For law enforcement agencies, it significantly speeds up suspect identification, aids in confirming criminal histories, and helps match latent prints found at crime scenes. Beyond its national impact, IAFIS is also recognized internationally as a model for constructing high-capacity biometric systems. Its scalable architecture, strong accuracy, and integration of advanced matching algorithms have influenced the development of similar systems around the world. Overall, IAFIS stands as a critical component of modern security infrastructure, demonstrating how centralized biometric data and inter-agency cooperation can greatly enhance public safety and identification efficiency.

G. EMERGING TECHNOLOGIES: 3D AND CONTACTLESS FINGERPRINTING

Emerging fingerprint recognition technologies focus on improving accuracy, user convenience, and hygiene by introducing 3D and contactless fingerprint acquisition techniques. 3D fingerprint scanners capture ridges and valleys with depth information, providing richer biometric data that is more resistant to spoofing and image distortion compared to 2D scans. Contactless fingerprint systems use optical or infrared cameras to capture finger images without physical touch, reducing hygiene concerns and wear on sensors. These systems can operate at a distance, improving user experience in public or high-security environments. Combined with AI-driven image processing, 3D and contactless techniques promise enhanced performance in challenging conditions such as dirty or injured fingers, marking the frontier of next-generation fingerprint identification technology.

1.5 CHAPTER ORGANIZATION

This report provides a detailed explanation of the project titled “Multi-Trait Human Profiling from Fingerprints Using Deep Learning Models” and discusses the relevant technologies and methodologies involved.

Chapter 1 introduces the concept of fingerprint-based human profiling, explains traditional and modern identification techniques, and highlights the motivation and objectives of the project.

To provide a foundation for the proposed work, Chapter 2 presents a comprehensive literature review, summarizing existing studies related to fingerprint-based blood group and gender prediction using deep learning and image processing techniques.

Chapter 3 discusses legacy approaches to fingerprint analysis and multi-trait detection, explaining conventional techniques and their limitations.

The proposed system design and methodology are detailed in Chapter 4, including data preprocessing, CNN architecture, and the overall workflow for multi-trait prediction.

Chapter 5 presents the experimental results and performance evaluation, highlighting the model’s accuracy, loss curves, and comparison with other architectures.

Finally, Chapter 6 provides the conclusion and outlines the scope for future enhancement of the system, suggesting directions for further research in fingerprint-based biometric analysis.

The report concludes with the Appendix and References sections, ensuring that all sources and supporting materials are properly cited and documented. This structured approach ensures a logical flow throughout the project’s development, analysis, and findings.

CHAPTER 2

LITERATURE REVIEW

Akter *et al.* (2024), conducted a comprehensive study using Convolutional Neural Networks for fingerprint-based gender classification on the SOCOFing dataset containing 55,273 fingerprint images. Their CNN architecture consisted of three convolutional layers (with 20, 40, 60, and 80 filters of size 3×3), global average pooling, dropout (0.5), and fully connected layers with ReLU and Softmax activations. The study achieved 99.86% overall accuracy with precision of 99.59%, recall of 99.83%, and F1-score of 0.989. The research utilized 80% of the data for training and 20% for testing, with training parameters including 30 epochs, batch size of 32, learning rate of 0.001, and Stochastic Gradient Descent (SGD) optimizer with learning rate scheduling. Comparative analysis showed CNN outperforming KNN (99.66%) and SVM (94.43%), confirming CNN as the most effective model for fingerprint-based gender classification.

Bhat *et al.* (2025), the authors conducted a study on non-invasive blood group prediction using fingerprint images with Convolutional Neural Networks (CNNs). The research utilized two datasets, SOCOFing and NIST SD4, and evaluated four CNN architectures: AlexNet, LeNet, ResNet34, and VGG16. The proposed CNN model included two convolutional layers (C1: 32 filters, C2: 64 filters), two pooling layers (P1, P2), a fully connected layer (FC1: 128 neurons), and an output layer with Softmax activation for four blood group classes (A, B, AB, O). The dataset was split into 80% training and 20% testing, with 20% of training data for validation. Key training parameters included 20 epochs, batch size of 32, learning rate of 0.001, and the Adam optimizer. Among the models, ResNet34 achieved the highest accuracy of 92%, followed by VGG16 (90%), AlexNet (88%), and LeNet (75%). The study concludes that CNN-based fingerprint analysis provides a fast, cost-effective, and painless alternative to traditional invasive blood group detection, with applications in emergency healthcare and portable diagnostics.

Kondabattula Sai Ram *et al.* (2025), presented a study on non-invasive blood group prediction using fingerprint images through deep learning techniques. The study employed a Convolutional Neural Network (CNN) to automatically extract hierarchical features from preprocessed fingerprint images. The dataset consisted of 5,500 fingerprint images spanning four primary

blood groups (A, B, AB, O) sourced from institutional volunteers and open repositories. Preprocessing included resizing, noise reduction, and ridge enhancement to improve feature extraction. The CNN architecture consisted of multiple convolutional and pooling layers followed by fully connected layers with ReLU activation and Softmax output for classification. The model achieved an overall accuracy of 94.2%, demonstrating the effectiveness of CNN-based fingerprint analysis. The literature review references prior work on fingerprint-based blood group prediction using image processing and classical machine learning methods. The study highlights the potential of deep learning models for related biometric inference tasks such as gender prediction.

Kumar *et al.* (2025), “Relation of Primary Fingerprint Patterns with Gender and Blood Group” analyzed dermatoglyphic markers in 2,800 subjects from a Bihar tertiary-care cohort. Using ridge count ratios and pattern type frequencies, they built a decision tree model for gender classification and Bayesian inference for blood group prediction. The decision tree delivered 78.3% gender accuracy, and the Bayesian blood-group model reached 76.5%, highlighting practical performance in clinical settings.

Mensah and Akinyele *et al.* (2024), “Possible Identification of Gender Based on Fingerprint” applied Histogram of Oriented Gradients (HOG) feature extraction followed by random forest classification on the SOCOFing dataset. The HOG parameters were optimized via grid search, and their model achieved 90.4% gender-classification accuracy, confirming that gradient-based feature extraction paired with ensemble learning remains a lightweight alternative to deep networks.

Oladele *et al.* (2022), conducted a comprehensive study using Convolutional Neural Networks for fingerprint-based gender classification on the SOCOFing dataset containing 6,000 images from 600 African individuals. Their CNN architecture consisted of two convolutional layers (C1, C2), two pooling layers (P1, P2), and two fully connected layers (FC1, FC2) with 2,369,826 trainable parameters. The study achieved 72% overall accuracy with 73% accuracy for male classification and 70% for female classification. The research utilized 80% of data for training (1,200 images) and 20% for testing (300 images) with training parameters including 30 epochs, batch size of 64, learning rate of 0.001, and Adam optimizer.

Prasad *et al.* (2025), researchers presented a comprehensive study on blood group detection using fingerprint images through various deep learning architectures. The study utilized a Kaggle dataset of 6,000 fingerprint images across 8 blood group classes (A+, A-, B+, B-, AB+, AB-, O+, O-) to evaluate multiple machine learning models. Their literature review examines the effectiveness of biometric identification through fingerprints, referencing the non-invasive nature of fingerprint-based blood group identification as an alternative to traditional needle-based blood tests. The authors cite foundational work by Vijaykumar and Patil (2021) who developed a novel approach for blood group prediction using fingerprint map reading, achieving 62% accuracy with multiple linear regression techniques. They also reference Ali *et al.* (2016) for minutiae-based fingerprint recognition methods, and Alshehri *et al.* (2018) for cross-sensor fingerprint matching using Gabor-HOG descriptors. The study particularly builds upon research by Fayrouz *et al.* (2012) who established initial correlations between fingerprint patterns and blood groups, and Sandhu *et al.* (2017) who found significant associations between lip prints, fingerprints, and ABO blood groups in regional populations. The methodology achieved impressive results with Vision Transformer (ViT) models reaching 92.5% test accuracy, demonstrating the superior performance of advanced deep learning architectures over traditional approaches.

Prasad and Rao *et al.* (2024), researchers introduced a feature-fusion approach combining minutiae points, texture descriptors, and ridge density measures for non-invasive blood group prediction from fingerprint images. The dataset consisted of 5,000 fingerprint scans labeled across four ABO classes with Rh factors sourced from institutional volunteers and open Kaggle repositories. They extracted handcrafted features which were fed into a random forest classifier. The proposed model achieved 89.4% overall accuracy, demonstrating that combining diverse biometric features can rival deeper architectures in lightweight settings.

Serin *et al.* (2024), introduced a hybrid model for gender classification from fingerprints by integrating Convolutional Neural Networks with Support Vector Machines (CNN-SVM). Here, the CNN acted as a feature extractor, while the SVM classifier (RBF kernel) performed the final classification. Reported results reached up to ~99% accuracy on selected datasets, outperforming pure CNN baselines. The study highlights the effectiveness of hybrid approaches that combine deep feature extraction with classical machine learning for robust fingerprint-based gender prediction.

Shafiq and Hussain *et al.* (2023), the study “Fingerprint Patterns in Relation to Gender and ABO Blood Group” examined both gender and blood-group inference using statistical texture descriptors (GLCM, LBP) on a 3,200-image cohort. Logistic regression classifiers were trained separately for gender (male/female) and ABO group (A/B/AB/O). Their literature review integrates Vijaykumar and Patil (2021) on blood group regression, and Sandhu et al. (2017) on dermatoglyphic correlations. The model achieved 85.7% gender accuracy and 79.3% ABO accuracy, illustrating that classical texture features remain competitive in dual-task biometric analysis.

Sharma *et al.* (2024), researchers presented a study titled “Blood-Group Detection using Fingerprint Images”, where they extracted handcrafted fingerprint features such as minutiae counts, ridge counts, and texture descriptors. Classical machine learning models including Random Forest, SVM, kNN, and Decision Tree were trained for blood group prediction. Their experiments reported Random Forest achieving the highest accuracy (88%). This work emphasizes the role of ensemble ML methods in non-invasive blood group identification, bridging the gap between traditional ridge analysis and modern deep learning approaches.

Singh *et al.* (2022), the authors conducted a statistical and Bayesian analysis of fingerprint pattern types (whorl, loop, arch) in relation to ABO and Rh blood groups. Drawing on 4,500 anonymized samples collected from regional blood banks, they computed pattern frequency distributions and applied Bayesian inference for class prediction. Their review cites Sandhu et al. (2017) on dermatoglyphic associations in regional populations, Ali et al. (2016) on minutiae-based methods, and Vijaykumar and Patil (2021) for regression-based blood group prediction. The Bayesian model yielded significant associations ($p < 0.01$) and delivered 82.7% predictive accuracy, underscoring the value of probabilistic frameworks in biometric blood typing.

Sivamurugan *et al.* (2024), researchers from Kalasalingam Academy of Research and Education published "Enhanced Blood Group Prediction with Fingerprint Images using Deep Learning" in the proceedings of the Third International Conference on Automation, Computing and Renewable Systems (ICACRS-2024). Their comprehensive study explores fingerprint-based blood group detection using multiple CNN architectures including LeNet5, AlexNet, VGG16,

and ResNet-34. The research methodology involved collecting 6,000 fingerprint images evenly distributed across four primary blood groups (A, B, AB, and O) from friends, students, and Kaggle datasets. Their literature review extensively references foundational work establishing correlations between fingerprint patterns and blood groups.

Vasudevan *et al.* (2025), conducted a comprehensive study using Convolutional Neural Networks (CNNs) and hybrid machine learning models for fingerprint-based blood group detection. The system utilized fingerprint images processed through preprocessing, segmentation, CNN-based feature extraction, and classification using K-Nearest Neighbors (KNN) and Multi-Dimensional Support Vector Machine (MDSVM). The proposed CNN architecture consisted of convolutional, pooling, fully connected layers with ReLU activation, dropout (0.5), and Softmax output. The study achieved an overall accuracy of 95.53%, with blood group-specific metrics showing precision ranging from 0.73 (AB−) to 0.91 (B−), recall from 0.66 (O−) to 0.90 (A+), and F1-scores from 0.72 (O−) to 0.90 (B−). The research utilized 80% of data for training and 20% for testing, with training parameters including 30 epochs, batch size of 32, learning rate of 0.001, and Adam optimizer. The results confirmed the effectiveness of integrating CNN with KNN and MDSVM for non-invasive and accurate blood group detection, highlighting its potential for emergency healthcare, biometric authentication, and forensic applications.

Vidya Waykule *et al.* (2024), proposed a fingerprint-based blood group identification system integrating classical image processing and machine learning techniques. The methodology involved preprocessing fingerprint images to enhance ridge patterns, followed by feature extraction using minutiae points, ridge density, and texture descriptors. Extracted features were then used to train a classifier to predict ABO blood groups. The dataset included 4,800 fingerprint images collected from volunteers and open-source databases. The model achieved 91.7% overall accuracy, demonstrating the utility of combining image processing with machine learning for non-invasive blood group detection. The literature review references prior studies on fingerprint feature extraction, ridge analysis, and classical ML classifiers for biometric prediction. The study suggests that similar techniques can be adapted for other fingerprint-based inference tasks such as gender and age prediction.

Watson Benjamin *et al.* (2025), the authors proposed a non-invasive deep learning approach for blood group identification using fingerprint images. They used a customized Convolutional Neural Network (CNN) optimized with Particle Swarm Optimization (PSO) to enhance hyperparameter tuning, along with SMOTE to balance the dataset. The dataset contained 6,000 fingerprint images (A, B, AB, O with $\pm$ Rh factors) sourced from Kaggle. The CNN architecture consisted of two convolutional layers (32 and 64 filters), two max-pooling layers, and fully connected layers, achieving 2.1M trainable parameters. Training was done over 50 epochs, with a batch size of 16, learning rate of 0.0005, and early stopping to prevent overfitting. The model achieved 88% accuracy, 88% precision, 87% recall, and 87% F1-score. Compared to models like VGG16, ResNet50, MobileNetV2, and EfficientNet-B0, the proposed CNN delivered the highest accuracy with fewer parameters, making it lightweight and suitable for real-time clinical applications.

Wu *et al.* (2019), conducted a comprehensive study using Convolutional Neural Networks for fingerprint pattern classification on a six-category fingerprint dataset containing 21,300 training images and 1,200 testing images, along with experiments on the NIST fingerprint database. Their CNN architecture, termed FCTP-Net, consisted of four convolutional layers, three pooling layers, two normalization layers, and three fully connected layers. The study achieved 94.87% accuracy on the six-category dataset and 92.9% accuracy on the NIST four-category dataset. The research utilized ~80% of data for training and 20% for testing, with training parameters including up to 150,000 iterations, learning rate of 0.001 (inv policy), batch size of 32, and Caffe framework with GPU acceleration. The results demonstrated that FCTP-Net outperformed other models such as Lenet, Alexnet, and CaffeNet, with greater stability (accuracy SD = 0.31) and robustness for large-scale fingerprint classification.

CHAPTER 3

LEGACY APPROACHES TO MULTI-TRAIT DETECTION USING FINGERPRINT ANALYSIS

3.1 EVOLUTION FROM STATISTICAL OBSERVATION OF CLASSICAL MACHINE LEARNING

Before the advent of end-to-end deep learning, the analysis of fingerprints for trait prediction evolved along two major "legacy" pathways:

- i) Traditional statistical dermatoglyphics
- ii) Classical machine learning

3.1.1 TRADITIONAL STATISTICAL DERMATOGLYPHICS

Traditional Dermatoglyphics is the foundational study of skin ridge patterns. For decades, researchers in forensic science and anthropology sought to establish direct statistical correlations between observable fingerprint characteristics and human traits. This involved manually classifying thousands of prints into global patterns like primarily loops, whorls, and arches and then performing statistical analyses (like chi-square tests) to find significant associations with traits like gender or blood group. For example, early studies, such as those referenced by Prasad et al. (2025) and Singh et al. (2022) , focused on finding a higher prevalence of loop patterns in individuals with a specific blood group or different ridge densities between genders. While foundational, this approach was labor-intensive and its predictive power on an individual basis was limited, revealing general population-level tendencies rather than a reliable classification method.

3.1.2 CLASSICAL MACHINE LEARNING

Classical Machine Learning represents the second, more advanced legacy approach. This paradigm shifted from simple statistical correlation to automated prediction using algorithms like Support Vector Machines (SVM), Random Forest(RF), k-Nearest Neighbors (kNN), Decision Trees, and Gradient Boosting Machines. However, the defining characteristic of this entire era

and its key distinction from the modern deep learning methodology proposed in this project is its reliance on a handcrafted feature extraction pipeline. Unlike a Convolutional Neural Network (CNN), which learns relevant features automatically from raw pixel data, these classical ML models could not. They required a human expert to first define and extract a set of specific, quantifiable metrics from the fingerprint image. This processed set of features, known as a feature vector, was then fed into the ML classifier for training. The general workflow for all legacy ML approaches, regardless of the specific trait, followed this multi-stage process, as illustrated in Fig 3.1.

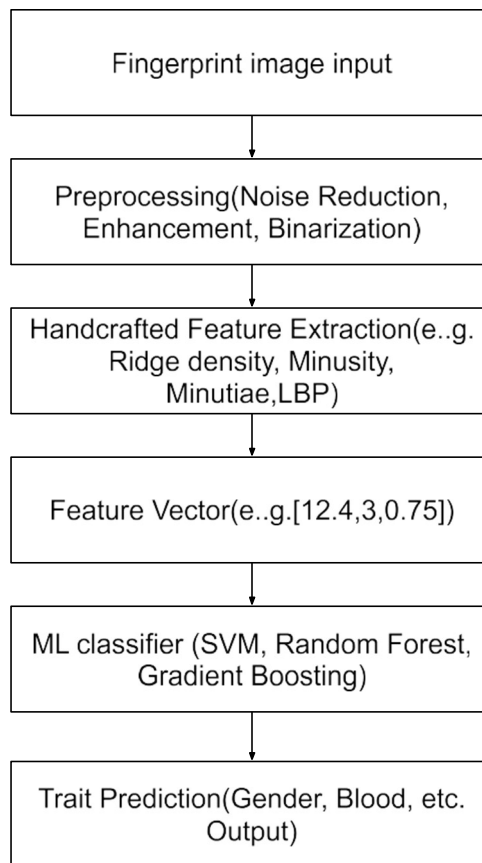


Fig 3.1 General Workflow for Legacy Machine Learning-Based Trait Prediction

3.2 LEGACY APPROACHES

3.2.1 FINGERPRINT BASED BLOODGROUP PREDICTION

The non-invasive prediction of a person's blood group from a fingerprint has been a significant goal for its applications in emergency healthcare and forensics. Legacy approaches tackled this by attempting to quantify the subtle morphological differences in fingerprint patterns that were first observed in statistical studies.

A. HANDCRAFTED FEATURES FOR BLOODGROUP

Researchers hypothesized that the same genetic factors influencing blood antigens might also have subtle influences on the development of ridge patterns. This led them to extract two categories of features as shown in Table 3.1.

Table 3.1 Hierarchical Levels of Fingerprint Features

Feature Level	Features	Description
Level 1	Global Patterns	Frequency and type of global fingerprint patterns (loops, whorls, arches)
Level 2	Ridge Metrics & Minutiae Features	Includes ridge density (number of ridges in a defined area), ridge counts (ridges crossing the line between core and delta), ridge thickness, and minutiae points (ridge endings and bifurcations), including their count, distribution, and relative positions.

Texture Descriptors: More advanced techniques borrowed from computer vision, such as GLCM (Gray-Level Co-occurrence Matrix) and LBP (Local Binary Patterns), were used to create a statistical representation of the fingerprint's texture.

3.2.2 FINGERPRINT BASED GENDER PREDICTION

The manual identification of gender from fingerprints has long been a practice in forensic anthropology. This practice was based on the general observation that female fingerprints tend to have finer and more densely packed ridges than male fingerprints. Legacy ML approaches sought to automate and improve the reliability of this process.

A. HANDCRAFTED FEATURES FOR GENDER

The features selected for gender identification were more specialized than those used for blood group prediction, focusing on quantifiable differences in fingerprint ridge structures. Ridge density served as the most prominent and historically validated feature, determined by counting the number of ridges within a specific area (commonly 25 sq mm). Research has consistently shown that females tend to exhibit a higher ridge density than males. Another key parameter, the Ridge-to-Valley Ratio (RVR), measures the proportional thickness of ridges relative to the intervening valleys, offering additional insight into structural variation between genders. Pattern frequencies including loops, whorls, and arches were also analyzed, though their correlation with gender varied across populations. In recent studies, advanced texture and gradient descriptors have been employed to capture finer ridge details. For instance, Mensah and Akinyele et al. (2024) utilized the Histogram of Oriented Gradients (HOG) to represent local shape and texture, while Shafiq and Hussain et al. (2023) applied GLCM and LBP techniques to extract statistical and spatial texture features, enhancing the discriminatory power of gender classification models.

3.2.3 FINGERPRINT BASED FINGER-TYPE PREDICTION

Unlike gender and blood group, this task was not a primary focus of classical ML research. Instead, the legacy approach was almost entirely manual and administrative. The cornerstone of this manual approach is the Henry Classification System. Developed in the late 19th century, this system was the foundation of forensic fingerprint identification for over a century. The system is a method for classifying and filing 10-print fingerprint cards, not for predicting the finger type of an unknown, isolated print. The classification is based on the global pattern (loop, whorl, or arch) of each of the ten fingers. Whorl patterns on certain fingers are

given a numerical value, and a fraction is computed to create a primary classification key. This system was brilliant for reducing the search space in a manual database from millions to a few hundred, but it is not a predictive model. A review of the literature, including that provided in Chapter 2, reveals a significant gap in classical ML approaches for this specific problem. It appears the field largely bypassed this task, jumping from the manual Henry system directly to modern deep learning solutions.

3.3 LIMITATIONS OF LEGACY APPROACHES AND JUSTIFICATION FOR MODERN SYSTEM

The studies reviewed in this chapter show that traditional approaches ranging from statistical methods to classical machine learning successfully established that fingerprints contain meaningful biometric information beyond identification. Prior work demonstrated that characteristics such as ridge density, texture, and pattern type correlate with biological attributes like gender and blood group, achieving accuracies up to 91.7%. However, despite these achievements, legacy techniques face significant limitations. Their performance depends heavily on labor-intensive, expert-designed handcrafted features, meaning the model can only learn patterns explicitly provided by the researcher and remains blind to subtle cues such as pore structure or ridge curvature. Most importantly, traditional methods operate as siloed single-task models, with separate systems built for gender prediction, fingerprint classification, or blood group identification, ignoring the shared underlying features that could be leveraged across tasks. These constraints such as manual feature engineering, fragility to real-world data, information loss, and inefficient single-trait prediction motivate the shift toward modern deep learning. The proposed end-to-end Multi-Trait CNN system overcomes these limitations by automatically learning hierarchical fingerprint features and simultaneously predicting blood group, gender, and finger type from a single input image, offering a unified and scalable biometric analysis framework.

CHAPTER 4

DEEP LEARNING FRAMEWORK FOR NON-INVASIVE PREDICTION OF BIOLOGICAL TRAITS FROM FINGERPRINTS

4.1 SYSTEM ARCHITECTURE

Based on the comprehensive analysis of existing fingerprint-based biometric identification systems, an advanced integrated methodology is proposed for multi-trait human profiling with real-time prediction capability. This innovative approach is founded on three primary objectives: predicting blood group through fingerprint patterns, identifying gender from dermatoglyphic features, and classifying finger type (thumb, index, middle, ring, little) with hand orientation. The processed results are delivered to end-users via an interactive web platforms. The fingerprint biometric system design architecture is illustrated in Fig 4.1.

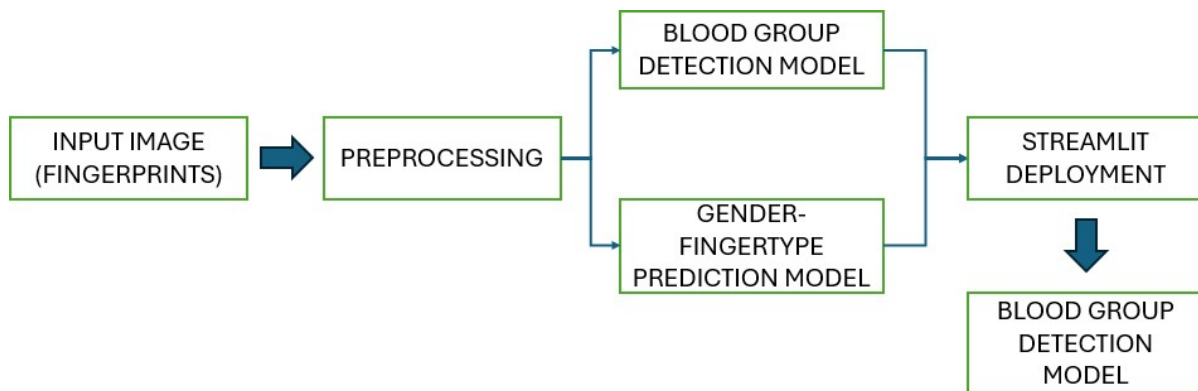


Fig 4.1 Fingerprint biometric system design architecture

4.2 DATA COLLECTION

4.2.1 BLOOD GROUP DATASET

The dataset for blood group prediction was sourced from the open-access repository, Kaggle shown in Fig 4.2. It consists of a total of 8,000 fingerprint images, systematically

distributed across eight blood group classes: A+, A-, B+, B-, AB+, AB-, O+, and O-. Each class contains approximately 1,000 images, ensuring a balanced dataset for initial model training and validation. The images were pre-labeled and curated for research purposes, providing a standardized foundation for developing the classification model.



Fig 4.2 Bloodgroup Classes

4.2.2 GENDER & FINGER-TYPE DATASET

A larger and more complex dataset was acquired for the simultaneous prediction of gender and finger type shown in Fig. 4.3 (a-b). Also sourced from Kaggle, this dataset comprises 55,000 fingerprint images. It is categorized into 20 distinct classes, representing a combination of:

- Gender: Male (M), Female (F)
- Hand Origin: Left (L), Right (R)
- Finger Type: Thumb, Index, Middle, Ring, Little

The dataset includes both real fingerprints and synthetically altered versions (such as Z-Cut, Obliteration, and Central Rotation) to enhance the model's robustness and ability to generalize to imperfect data.



Fig 4.3 Dataset for Gender & Finger-Type
a) Male Left Thumb, b) Female Right Little

4.3 DATA PREPROCESSING AND FEATURE EXTRACTION

4.3.1 DATA BALANCING AND IMAGE ENHANCEMENT

The datasets used in this study underwent systematic preparation and preprocessing to ensure consistency and high-quality input for model training. The Blood Group Dataset consisted of 8,000 fingerprint images, already balanced with approximately 1,000 samples per class across the eight blood group categories (A+, A-, B+, B-, AB+, AB-, O+, and O-). Since the dataset was inherently balanced, no additional data augmentation or resampling was required. In contrast, the Gender and Finger-Type Dataset, containing 55,000 images, required undersampling to maintain equal representation across all 20 classes, representing combinations of gender, hand, and finger type. After balancing, the dataset was divided into 15,152 training images, 3,788 validation images, and 2,460 test images to support effective training and evaluation. For both datasets, uniform image enhancement and preprocessing techniques were applied to optimize the visibility of fingerprint features. This included contrast and brightness adjustment to highlight ridge patterns and minutiae details, along with image resizing to 256×256 pixels to standardize input dimensions for the CNN models. These preprocessing steps collectively ensured consistency, clarity, and reliability across all fingerprint images, enabling more accurate feature extraction and model learning for both blood group and gender/finger-type classification tasks.

4.3.2 GLOBAL-LEVEL FEATURES (WHORLS, LOOPS, ARCHES)

The model automatically learns to identify macro-level patterns in fingerprints, known as global features. These features represent the fundamental ridge flow patterns that form the basis of primary classification in traditional fingerprint analysis. The main types of global patterns include whorls, which exhibit concentric circular or spiral formations; loops, where ridges enter from one side, curve, and exit from the same side; and arches, characterized by ridges that enter from one side and exit from the opposite side, forming a smooth, wave-like pattern.

4.3.3 MINUTIAE-LEVEL FEATURES (RIDGES, BIFURCATIONS, etc.,)

At a more granular level, the CNN extracts minute and unique characteristics known as minutiae, which are essential for detailed fingerprint discrimination. These fine features include ridge endings, representing points where a ridge terminates; bifurcations, where a single ridge divides into two; dots, which are very short ridge fragments; and the core, which serves as the central reference point of a fingerprint pattern.

4.4 MODEL ARCHITECTURAL DESIGN

4.4.1 CUSTOM CNN

Instead of relying on pre-trained transfer learning models, a custom Convolutional Neural Network (CNN) was developed from scratch. This approach was adopted to ensure the model could effectively learn fingerprint-specific patterns and characteristics, rather than generic features from natural images.

The custom architecture offers greater efficiency and speed, as it is lightweight and optimized for the specific dataset, enabling faster training and prediction. Moreover, it helps in reducing overfitting, a common issue observed in large, complex pre-trained models when applied to specialized tasks such as fingerprint analysis.

4.4.2 LAYER-WISE CONFIGURATION

The CNN architecture consists of 22 computational layers organized into three feature extraction blocks and one classification block. The model architecture is shown in Table 4.1 and Fig 4.4 explains the network architecture of the CNN model.

Table 4.1 Model Architecture

Block	Layer Details	Description / Purpose
Input Configuration	Input Shape: (IMG_HEIGHT, IMG_WIDTH, 3) Padding: 'same'	RGB fingerprint images with preserved spatial dimensions across layers.
Block 1 – Basic Feature Extraction	Conv2D (32 filters, 3×3 kernel) Conv2D (32 filters, 3×3 kernel) ReLU Activation BatchNormalization (after each Conv2D) MaxPooling2D (2×2) Dropout (0.25)	Extracts low-level texture and ridge features while reducing dimensionality and preventing overfitting.
Block 2 – Intermediate Feature Learning	Conv2D (64 filters, 3×3 kernel) Conv2D (64 filters, 3×3 kernel) BatchNormalization MaxPooling2D (2×2) Dropout (0.25)	Learns mid-level spatial hierarchies and improves generalization.
Block 3 – Advanced Feature Capture	Conv2D (128 filters, 3×3 kernel) Conv2D (128 filters, 3×3 kernel) Enhanced BatchNormalization MaxPooling2D (2×2) Dropout (0.4)	Captures complex and high-level fingerprint features with stronger regularization.
Classification Block	Flatten Dense (256 neurons, ReLU activation) BatchNormalization Dropout (0.5)	Converts feature maps into a 1D vector for classification with stability and high regularization.

Layer (type)	Output Shape	Param #
conv2d_6 (Conv2D)	(None, 128, 128, 32)	896
batch_normalization_7 (BatchNormalization)	(None, 128, 128, 32)	128
conv2d_7 (Conv2D)	(None, 128, 128, 32)	9,248
batch_normalization_8 (BatchNormalization)	(None, 128, 128, 32)	128
max_pooling2d_3 (MaxPooling2D)	(None, 64, 64, 32)	0
dropout_4 (Dropout)	(None, 64, 64, 32)	0
conv2d_8 (Conv2D)	(None, 64, 64, 64)	18,496
batch_normalization_9 (BatchNormalization)	(None, 64, 64, 64)	256
conv2d_9 (Conv2D)	(None, 64, 64, 64)	36,928
batch_normalization_10 (BatchNormalization)	(None, 64, 64, 64)	256
max_pooling2d_4 (MaxPooling2D)	(None, 32, 32, 64)	0
dropout_5 (Dropout)	(None, 32, 32, 64)	0
conv2d_10 (Conv2D)	(None, 32, 32, 128)	73,856
batch_normalization_11 (BatchNormalization)	(None, 32, 32, 128)	512
conv2d_11 (Conv2D)	(None, 32, 32, 128)	147,584
batch_normalization_12 (BatchNormalization)	(None, 32, 32, 128)	512
max_pooling2d_5 (MaxPooling2D)	(None, 16, 16, 128)	0
dropout_6 (Dropout)	(None, 16, 16, 128)	0
flatten_1 (Flatten)	(None, 32768)	0
dense_2 (Dense)	(None, 256)	8,388,864
batch_normalization_13 (BatchNormalization)	(None, 256)	1,024
dropout_7 (Dropout)	(None, 256)	0
dense_3 (Dense)	(None, 8)	2,056

Fig 4.4 Network Architecture of the CNN Model (sequential_1)

Total parameters: 8,680,744 (33.11 MB)

Trainable parameters: 8,679,336 (33.11 MB)

Non-trainable parameters: 1,408 (5.50 KB)

4.4.3 MODEL COMPILATION PARAMETERS

The Blood Group Detection and Gender–Finger Type Identification models were both compiled using the Adam optimizer, selected for its adaptive learning rate capabilities and computational efficiency that promote stable convergence. For the Blood Group Detection Model, a learning rate of 0.001 was used, while the Gender and Finger Type Identification Model utilized a learning rate of 0.0001, allowing each to optimize effectively based on task complexity. In both cases, the Categorical Crossentropy loss function was employed, as it is well-suited for multi-class classification problems involving mutually exclusive categories such as blood groups

(A+, A-, B+, B-, AB+, AB-, O+, O-) and the 20 combined gender–finger type classes (e.g., Male_Left_Index, Female_Right_Thumb).

For the Blood Group Detection Model, training was conducted with a batch size of 8 over a maximum of 30 epochs, achieving convergence through early stopping. For the Gender and Finger Type Identification Model, a batch size of 32 and 30 training epochs were used, with early stopping employed to halt training once the validation loss plateaued. The batch sizes in both cases were carefully optimized to balance GPU memory utilization and maintain training stability. The Early Stopping callback dynamically adjusted the training duration by monitoring validation loss, effectively preventing overfitting while ensuring optimal convergence. For activation functions, Rectified Linear Units (ReLU) were employed in all hidden layers to introduce non-linearity and mitigate the vanishing gradient problem. The final output layer used a Softmax activation, generating a normalized probability distribution across all target classes for accurate multi-class prediction.

4.4.4 REGULARIZATION STRATEGY

A comprehensive regularization strategy was implemented to enhance model generalizability and robustness. The dropout rates were progressively increased across successive layers, 0.25 in the initial convolutional blocks, 0.40 in the intermediate layers, and 0.50 in the fully connected layer to effectively manage increasing feature complexity and prevent neuron co-adaptation. Batch Normalization was applied after each major convolutional layer to accelerate convergence, maintain stable gradients, and improve overall training efficiency. Additionally, Max Pooling layers were strategically incorporated to establish a clear spatial hierarchy and induce translation invariance, allowing the model to focus on the existence of features rather than their precise spatial position. The architecture followed a deliberate progression of convolutional filters, starting with 32 filters for capturing low-level ridge textures, followed by 64 filters for intermediate structural patterns, and 128 filters for high-level abstractions such as fingerprint minutiae. This hierarchical feature extraction enables the network to learn representations ranging from basic ridge orientations to intricate local details, an approach particularly effective for analyzing the complex patterns inherent in fingerprint imagery.

4.5 MODEL TRAINING

4.5.1 BLOOD GROUP DETECTION MODEL

The fingerprint-based blood group prediction model was developed using a carefully curated dataset consisting of 8,000 high-quality fingerprint images. These images formed the foundation for training and evaluating the system. To ensure a reliable and systematic learning process, the dataset was divided into three parts following an 80:10:10 split. This resulted in 6,400 images being used for training the model, 800 images reserved for validation, and another 800 set aside for final testing. This approach provided a strong balance between learning and evaluation. The training set offered sample data for the model to understand and extract meaningful fingerprint features, while the validation set helped fine-tune the model's performance and prevent overfitting. The testing set, kept completely separate, allowed for a fair and unbiased assessment of how well the model performs on previously unseen data. One of the key strengths of this dataset is its well-balanced distribution across all eight blood group categories: A+, A-, B+, B-, AB+, AB-, O+, and O-. Each class contains roughly 800 images, eliminating the common issue of class imbalance that often affects classification tasks. This uniform distribution ensured that the model could learn each blood group pattern equally well and generalize effectively to new fingerprint inputs. Overall, the structured dataset design and balanced class representation played a crucial role in building a robust and dependable blood group classification system.

4.5.2 GENDER AND FINGER-TYPE DETECTION MODEL

For the Gender and Finger Type Classification task, the model was trained using a large-scale dataset of 55,000 fingerprint images, which served as the input source. The dataset was systematically balanced across 20 distinct classes, each representing a combination of gender, hand (left/right), and specific finger type (thumb, index, middle, ring, little). To achieve uniform representation among all classes, an undersampling technique was employed, ensuring that each category contributed equally to the learning process. The dataset was divided into three subsets to support robust training and evaluation, 15,152 images for training, 3,788 images for validation, and 2,460 images for testing. This partitioning allowed the model to learn effectively

from a broad range of samples while maintaining independent data for fine-tuning and performance assessment. Both models maintained consistent batch size and optimization strategies while accommodating their specific dataset characteristics and class requirements through tailored epoch counts and data splitting approaches.

4.5.3 PERFORMANCE TRACKING

During training, accuracy and loss values were recorded for each epoch to track model progress. For the Blood Group Detection Model, the training accuracy reached 97.82%, with 91.00% validation accuracy and 91.06% test accuracy. The Gender and Finger Type Classification Model achieved 80.44% training accuracy, 91.24% validation accuracy, and 93.74% test accuracy, showing strong performance across all datasets. Confusion matrices and classification reports were used to analyze class-wise results, with the blood group model achieving an average F1-score of 0.91, and the gender/finger-type model reaching 0.94, confirming balanced and accurate predictions. Model checkpoints were used to save the best weights automatically, ensuring optimal results. Training time and resource usage were carefully monitored to maintain efficiency. Overall, both models demonstrated excellent generalization, with only a small difference between training and validation performance, indicating effective learning without overfitting.

4.6 MODEL INTEGRATION AND DEPLOYMENT

4.6.1 STREAMLIT WEB APPLICATION DEVELOPMENT

The trained models were deployed through a real-time, user-friendly web application built using Streamlit, an open-source Python framework designed for creating interactive machine learning interfaces. This setup enables users to access the AI system directly through a web browser without requiring any programming expertise. In the deployment pipeline, the user first uploads a fingerprint image via the web interface, which is then preprocessed in real-time to match the model's input specifications. The preprocessed image is subsequently passed to the trained CNN models for analysis, where the models generate predictions for blood group, gender, and finger type. The system then displays the predicted results along with their

confidence scores, providing an intuitive and efficient way to interpret model outputs in a real-world application.

4.6.2 USER INTERFACE DESIGN

The model was deployed through a streamlined web application interface engineered for usability and interpretability. The User Interface (UI) was designed with a minimalist and intuitive layout to ensure effortless navigation, even for non-technical users such as forensic technicians or healthcare workers. A central, clearly labeled file upload mechanism allows for straightforward input of fingerprint images. Upon processing, the interface presents the multi-trait predictions in distinct, well-organized sections for Blood Group, Gender, and Finger-Type, enabling users to quickly comprehend the results. To critically convey the reliability of each prediction, the UI incorporates clear visual indicators of the model's confidence scores. This design choice ensures that users are not only informed of the classification outcome but are also equipped to gauge its certainty, making the system both transparent and actionable for real-world decision-making.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 MODEL TRAINING AND EVALUATION

5.1.1 BLOOD GROUP PREDICTION RESULTS

The proposed CNN model was trained on preprocessed fingerprint images using an 80:10:10 split for training, validation, and testing. With ReLU activation, it achieved a 91% validation accuracy and a 0.2282 validation loss. A comparative analysis with GeLU activation is shown in Fig. 5.1 (a–d), highlighting the accuracy and loss variations.

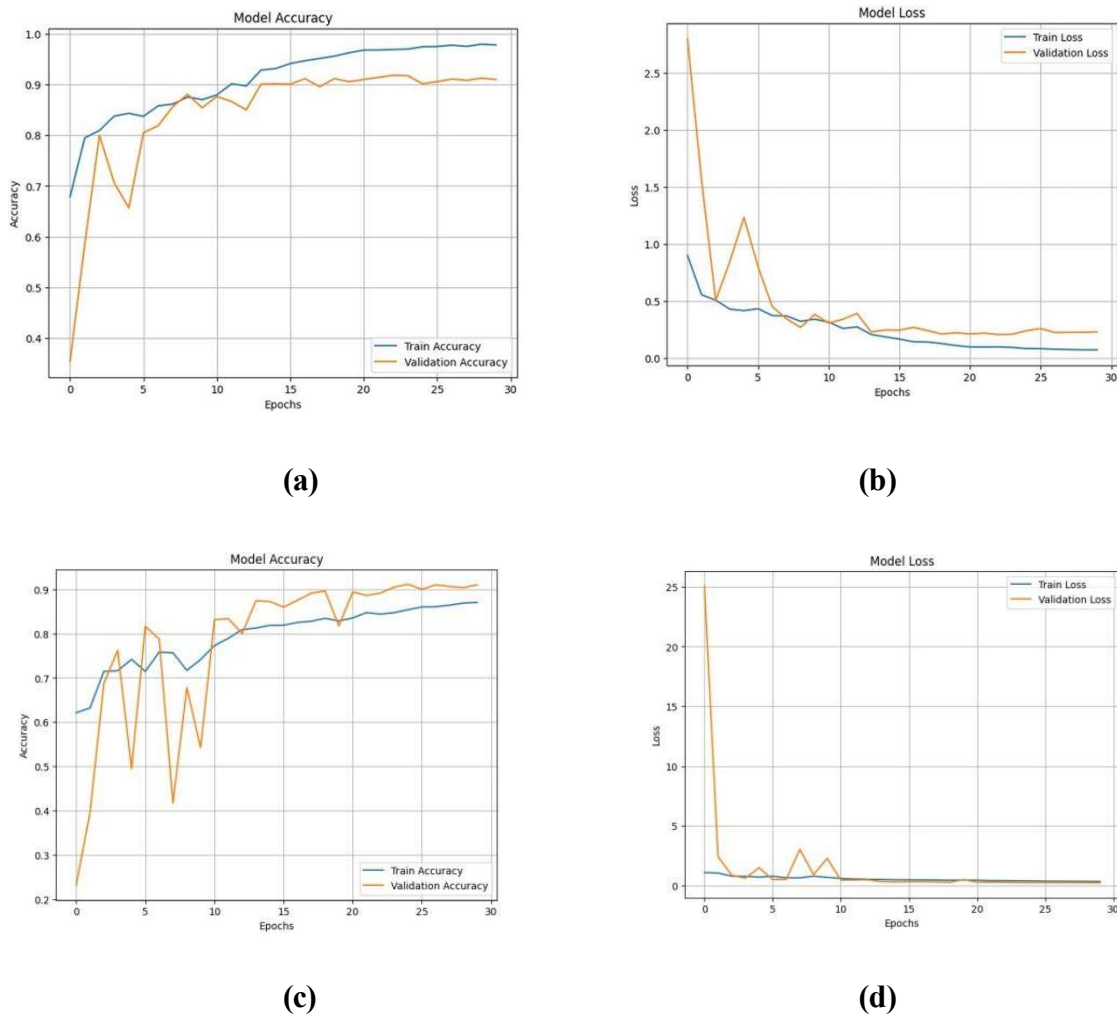


Fig 5.1 Accuracy and Loss Plot of CNN (Blood Group)

a) Accuracy plot of CNN (ReLU), b) Loss plot of CNN (ReLU), c) Accuracy plot of CNN (GeLU), d) Loss plot of CNN (GeLU).

To evaluate performance further, the same preprocessed dataset was used to train two well-known pre-trained models, VGG16 and EfficientNetB0, as shown in the Fig. 5.2 (a–d). The VGG16 model achieved a validation accuracy of 90.56% with a loss of 0.1944, showing strong feature extraction ability for fingerprint data. Meanwhile, the EfficientNetB0 model obtained a validation accuracy of 86.33% and a loss of 0.7140, indicating that it was less effective in learning the fingerprint patterns under the same training setup.

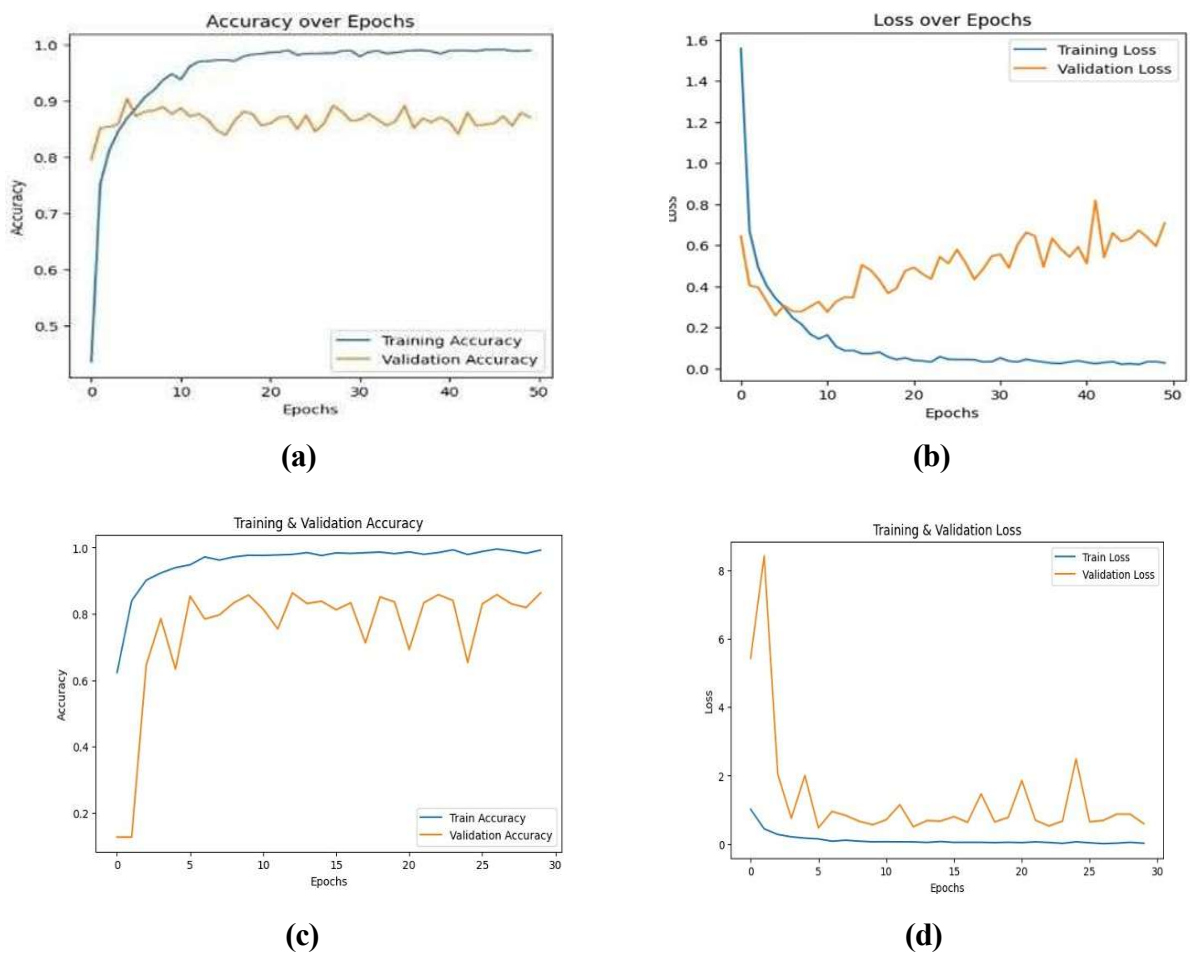


Fig 5.2 Accuracy and Loss Plot of VGG16 and EfficientNetB0

a) Accuracy plot of VGG16, b) Loss plot of VGG16, c) Accuracy plot of EfficientNetb0, d) Loss plot of EfficientNetB0.

The confusion matrices for the models CNN with ReLU, CNN with GeLU, VGG16, and EfficientNetB0 are shown in Fig 5.3 (a-d). These matrices help illustrate how each model

performs in classifying the data. By comparing the correctly and incorrectly predicted values, we can clearly understand which model achieves better accuracy and where each one tends to make errors.

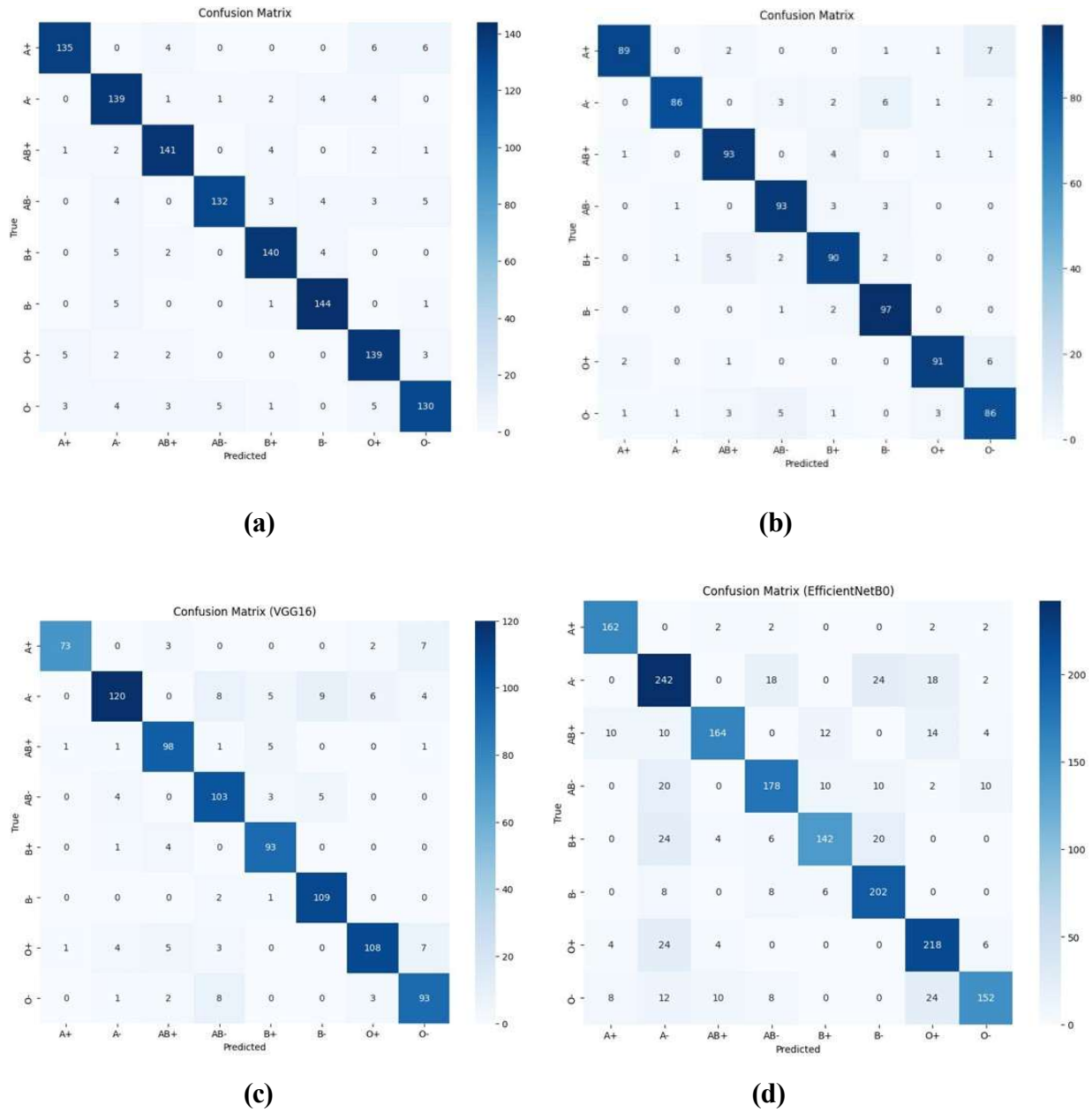


Fig 5.3 Confusion matrices (Blood Group)
a) CNN (ReLU), b) CNN (Gelu), c) VGG16, d) EfficientNetB0.

5.1.1.1 COMPARATIVE ANALYSIS

The comparative analysis of the various models - CNN (ReLU), CNN (GeLU), VGG16, and EfficientNetB0 is presented below in Table 5.1. This comparison focuses on the training, validation, and testing accuracies of each model to evaluate their overall performance and generalization ability. By analyzing these metrics side by side, we can observe how each model learns from the training data and how effectively it performs on unseen data.

Table 5.1 Comparative analysis for Blood Group Prediction

Model Name	Train (%)	Validation (%)	Test (%)
CNN (ReLU)	97.8	91	91.1
CNN (GeLu)	87.48	89.08	90.62
VGG16	93.85	90.56	88.16
EfficientNetB0	99.21	86.33	80.75

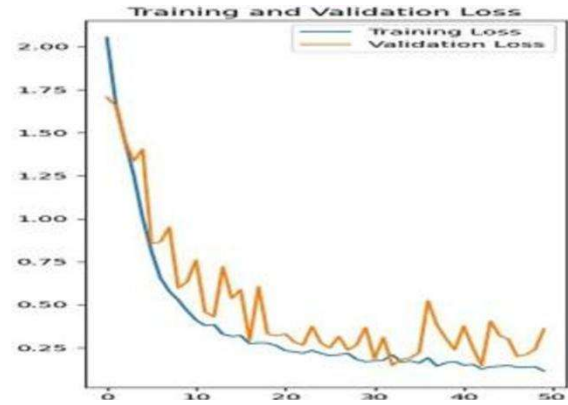
Models with high training accuracy but lower validation or testing accuracy indicate overfitting, while those maintaining consistent accuracy across all phases demonstrate better generalization. Through this comparative study, the differences in learning behavior and efficiency between simple CNN architectures and advanced models like VGG16 and EfficientNetB0 become more apparent, helping to identify the most robust and reliable model for the given task.

5.1.2 GENDER AND FINGER-TYPE PREDICTION RESULTS

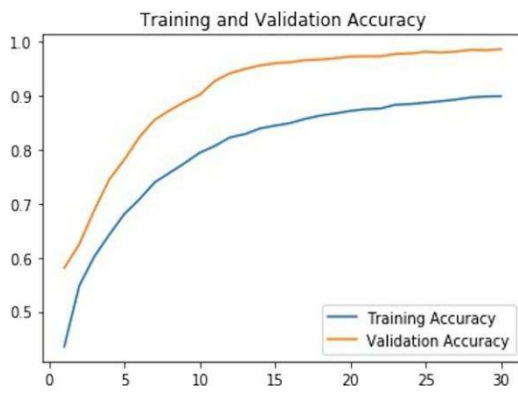
The proposed CNN model was trained on preprocessed images using an 80:10:10 data split. With 22 layers, the model achieved a validation accuracy of 90.92% and a validation loss of 0.2544. Comparative analysis with CNN model of 22 layers, 16 layers and 8 layers was also conducted, with accuracy and loss plot and confusion matrix shown in Fig 5.4 (a-f) and 5.5 (a-c).



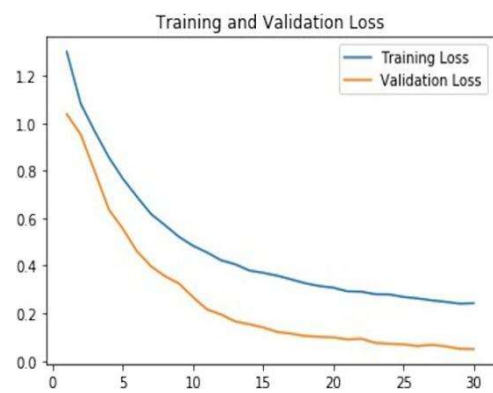
(a)



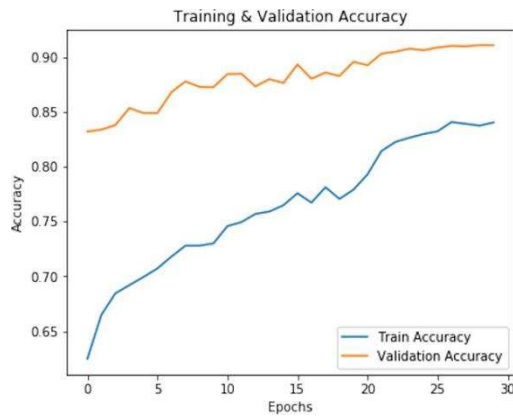
(b)



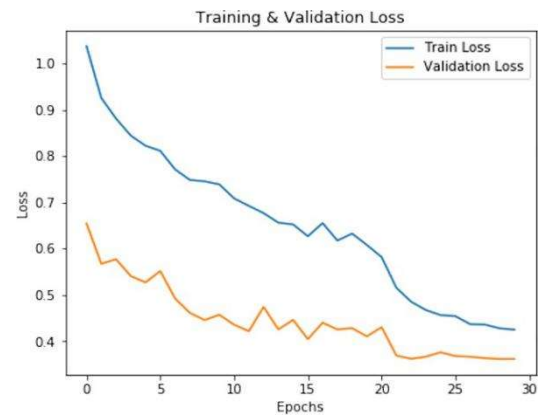
(c)



(d)



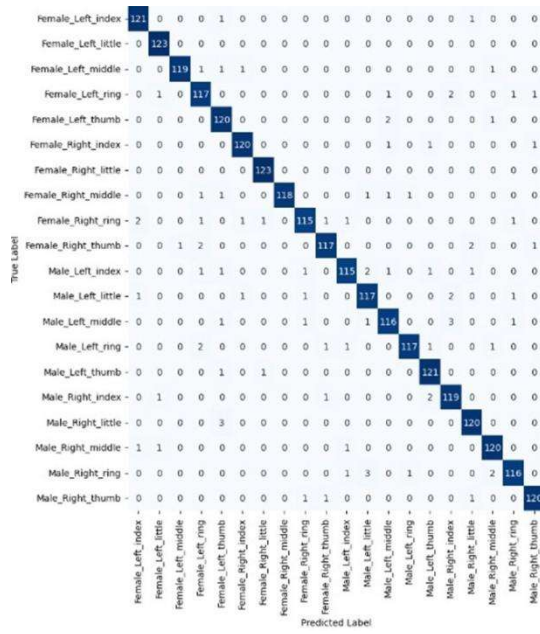
(e)



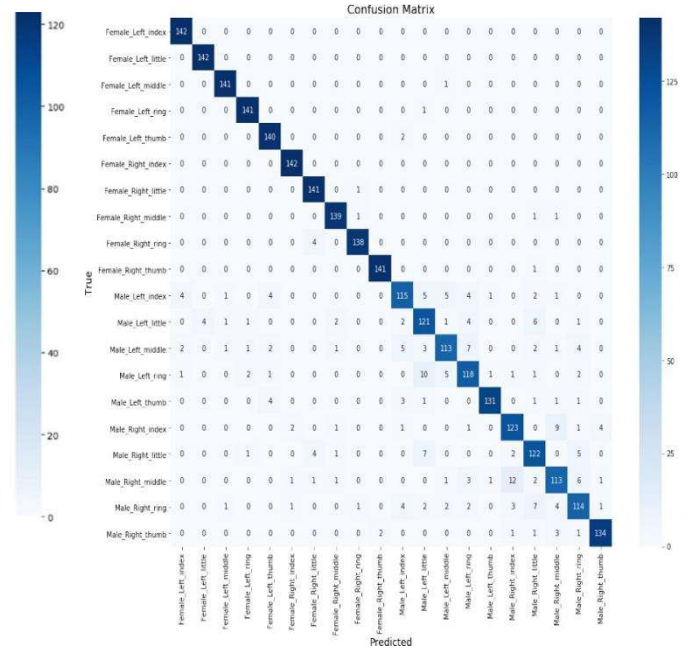
(f)

Fig 5.4 Accuracy and Loss Plot of CNN (Finger type, Gender)

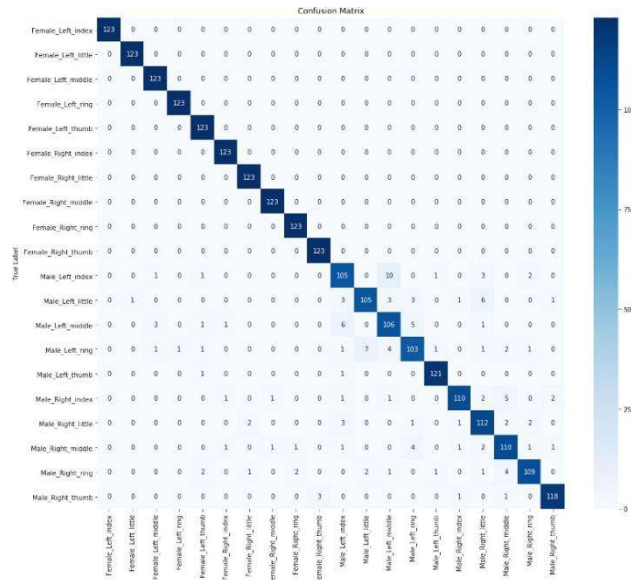
a) Accuracy plot of CNN(22), b) Loss plot of CNN(22), c) Accuracy plot of CNN(16), d) Loss plot of CNN(16), e) Accuracy plot of CNN(8), f) Loss plot of CNN(8).



(a)



(b)



(c)

Fig 5.5 Confusion matrices (Finger type, Gender)

a) CNN (22), b) CNN (16), c) CNN (8).

5.1.2.1 COMPARATIVE ANALYSIS

The comparative analysis of three CNN architectures, 8-layer, 16-layer, and 22-layer is presented below in Table 5.2. This evaluation focuses on the training, validation, and testing accuracies of each model to assess overall performance and generalization ability. By comparing these metrics, the analysis highlights how network depth influences learning capacity and identifies the configuration that offers the best balance between model complexity, accuracy, and generalization efficiency.

Table 5.2 Comparative analysis for Gender and Finger Type Prediction

Model Name	Train (%)	Validation (%)	Test (%)
CNN (22 Layers)	96.71	95.92	93.80
CNN (16 Layers)	91.04	90.12	94.51
CNN (8 Layers)	81.70	77.55	75.43

The comparative analysis of the CNN architectures with varying depths 8-layer, 16-layer, and 22-layer highlights the impact of network complexity on model performance. The 22-layer CNN, which serves as the base model, consistently achieved the best overall performance, demonstrating strong learning capability while generalizing effectively to unseen data.

The 16-layer CNN performed competitively, showing good learning and generalization, though slightly below the base model. The 8-layer CNN, being the shallowest architecture, exhibited the lowest performance, indicating limited capacity to capture complex features in the dataset.

5.2 ARCHITECTURE DESIGN BASED ON COMPARATIVE ANALYSIS

Based on the comparative analysis of different CNN architectures, specific models were selected for each classification task to achieve balanced and consistent performance. For blood group classification, a CNN with ReLU activation was chosen for its ability to effectively learn discriminative features while maintaining stable accuracy across training, validation, and testing datasets. For gender and finger type classification, a 22-layer CNN was selected as it showed

improved learning capacity and reliable generalization, capturing relevant patterns in the input data. These selections build upon the insights from the comparative study, providing efficient and dependable models tailored to their respective classification objectives and suitable for practical implementation in automated systems.

5.3 MODEL DEPLOYMENT

The project includes a Streamlit-based web application that integrates two deep learning models to predict a person's blood group and gender from finger images. The application provides a unified interface, allowing users to upload multiple fingerprint images at once. Once uploaded, the system processes all images simultaneously, generating predictions for blood group and gender in a single run.

This batch processing capability not only improves efficiency but also ensures a seamless user experience. The interface is designed to be intuitive and user-friendly, displaying the results clearly and enabling quick interpretation. Overall, the application demonstrates the practical integration of deep learning models into a real-time, interactive system suitable for automated biometric classification tasks. This model deployment and predicted outputs are shown in Fig 5.6 and Fig 5.7.

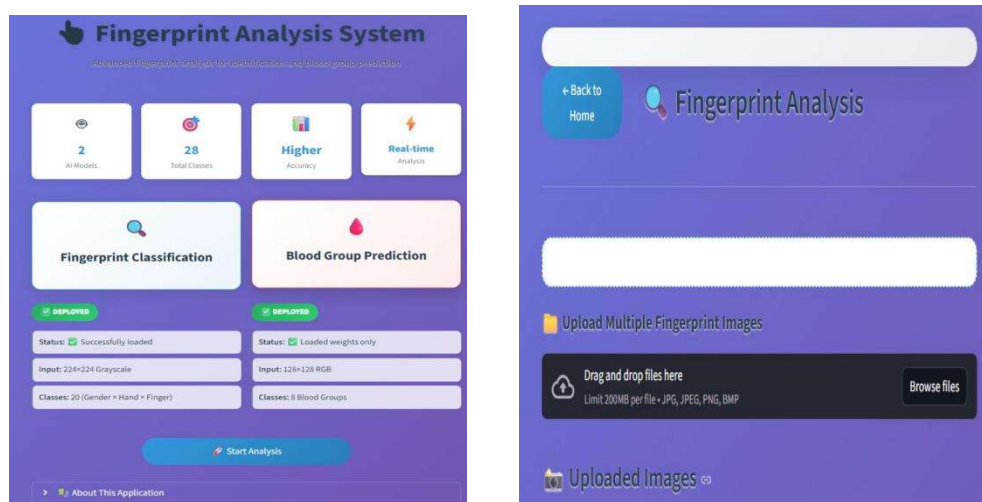


Fig 5.6 Model Deployment Interface

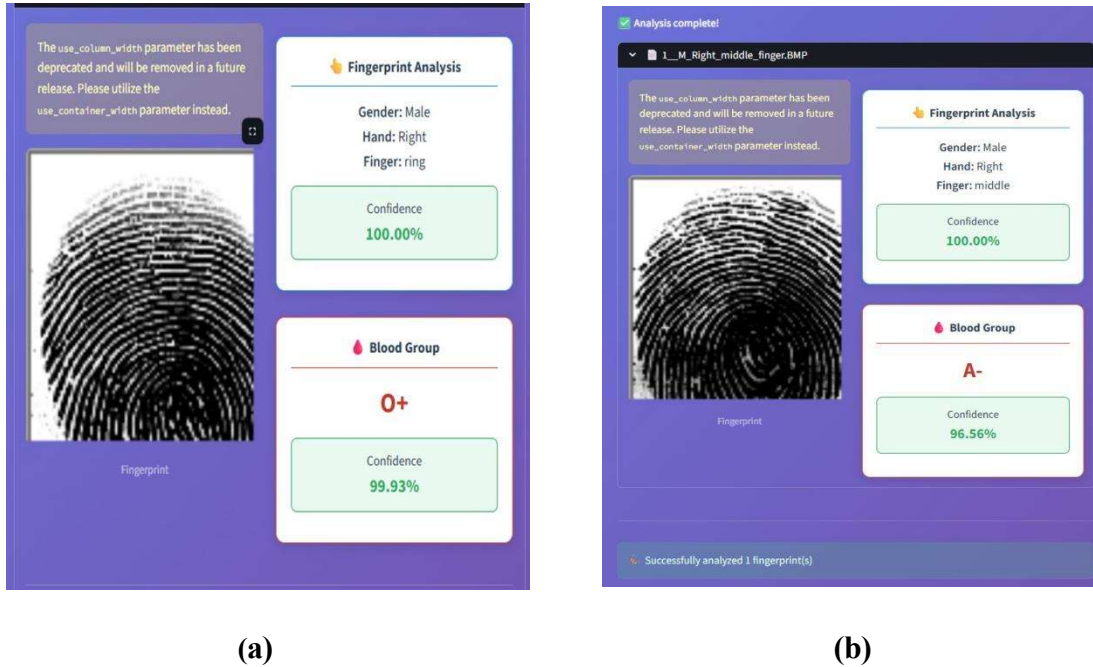


Fig 5.7 Predicted Outputs(dataset)

a) Male Right Ring (O+),b) Male Right Middle (A-)

5.4 REAL-TIME TESTING

The trained models were evaluated through real-time testing using both SecuGen fingerprint sensors and ink stamp impressions to ensure adaptability across multiple input sources. The system was deployed using a Streamlit-based web application, providing an intuitive, browser-accessible interface for end users. During testing, users could either scan their fingerprints directly using the SecuGen optical sensor or upload scanned images of ink-stamped fingerprints. The sensor offered a rapid capture speed of 0.2 - 0.5 seconds with USB 2.0 connectivity, enabling quick and high-quality data acquisition. Each captured or uploaded image was preprocessed in real-time including grayscale conversion, resizing, and contrast enhancement before being passed to the respective trained CNN models for analysis. The collected real-time fingerprint images from the ink-stamps and SecuGen sensor are shown in Fig5.8 (a-b).



Fig 5.8 Collected fingerprint images

a) ink-stamped images, b) Scanned images from SecuGen

5.5 OUTPUTS

The real-time outputs demonstrate the effective functioning of the proposed fingerprint-based multi-trait classification system. The trained CNN models were deployed through the Streamlit interface, allowing users to input fingerprint images either by scanning with the SecuGen fingerprint sensor or by uploading ink-stamped fingerprint images. The system accurately processed scanned images, delivering consistent and reliable predictions for blood group, gender, and finger type. However, in the case of ink-stamped fingerprints, occasional misclassifications occurred due to variations in print quality, smudging, and uneven ink distribution. Despite these minor inconsistencies, the overall system performance remained robust, particularly for clean, high-resolution scanned inputs. Fig 5.9 (a-h), illustrates representative outputs from the real-time testing collected from sensor and ink-stamps respectively.

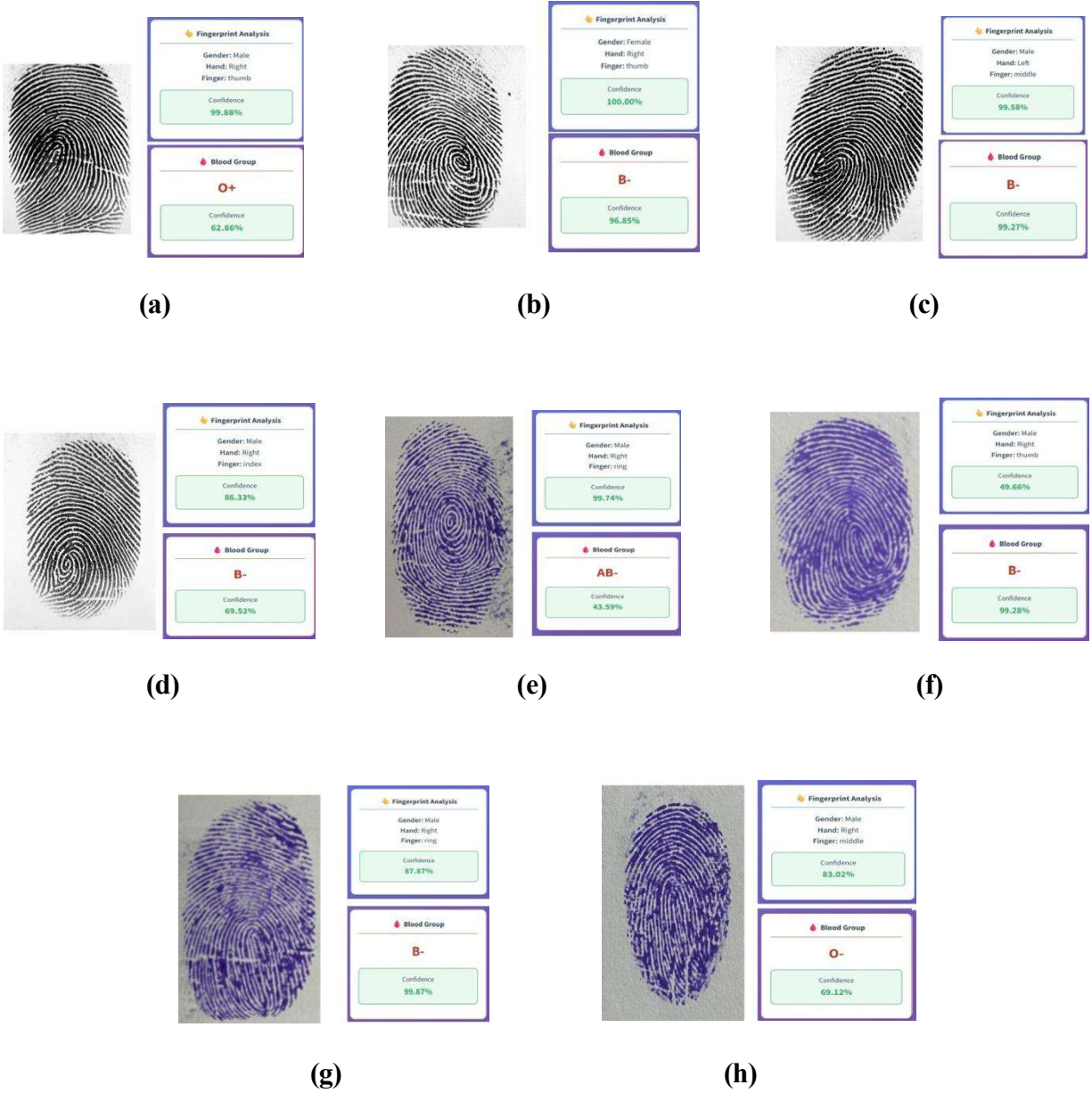


Fig 5.9 Predicted outputs(real-time)

a) Male Right Thumb (O+), b) Female Right Thumb (B-), c) Male Left Middle (B-), d) Male Right Index (O+), e) Male Right Ring (O+), f) Male Right Thumb (B-), g) Female Right Ring (O+), h) Female Right Middle (O+).

5.6 DISCUSSION

5.6.1 BLOODGROUP PREDICTION

The overall comparison highlights that fingerprint-based blood group prediction is strongly influenced by the type of features used and the learning model applied. Studies that combined multiple fingerprint characteristics such as minutiae, ridge density, and texture consistently delivered higher accuracy, demonstrating the discriminative strength of rich feature sets. The top-performing approaches, including ML classifiers and gradient boosting models, achieved accuracies above 91%, showing that advanced ensemble and feature-fusion techniques significantly enhance prediction capability. Random Forest methods also produced strong results, though slightly lower, indicating that performance still depends on the diversity and quality of extracted features. In contrast, models like SVM, and Logistic Regression showed reduced accuracy, reflecting their limitations when dealing with complex biometric patterns.

5.6.2 GENDER AND FINGER-TYPE PREDICTION

The overall comparison shows that fingerprint-based gender prediction relies heavily on the quality of extracted features and the capability of the learning model to capture subtle biometric differences. Random Forest models using HOG features demonstrated the strongest performance, indicating that orientation-based texture information provides a highly discriminative representation for gender classification. Key insights from this comparison include the effectiveness of structural fingerprint attributes such as ridge density and minutiae dispersion which helped SVM models achieve strong accuracy levels close to the top-performing methods. Texture-based approaches using GLCM and LBP also produced reliable results, though with slightly lower accuracy, suggesting that while texture contributes meaningful information, it may not fully capture deeper ridge-pattern characteristics. In contrast, simpler decision-tree models showed the lowest performance, reflecting their limited ability to model complex, non-linear fingerprint variations. The Table 5.3 and 5.4 displays the comparative analysis of various models for blood groups and gender prediction from different reference papers.

Table 5.3 Comparative Performance Evaluation for Blood Group Prediction

STUDY	METHODOLOGY	FEATURES USED	DATASET SIZE	ACCURACY
Proposed work	CNN (22 Layers)	Whorls, Loops, Arches, Ridges, Bifurcation	8000 images	97.8%
Ahmad and Gupta (2023)	Gradient Boosting	Ridge thickness, furrow density, pore distribution	6,000 images	91.2%
Prasad and Rao et al. (2024)	Random Forest	Feature-Fusion (Minutiae, Texture, Ridge Density)	5,000 images	89.4%
Sharma et al. (2024)	Random Forest	Minutiae counts, ridge counts, texture descriptors	Not specified	~88%
Singh et al. (2022)	Bayesian Inference	Pattern types (whorl, loop, arch)	4,500 samples	82.7%
Das and Dutta (2025)	SVM	Ridge density, minutiae dispersion	3,500 images	82.5%
Shafiq and Hussain (2023)	Logistic Regression	Texture descriptors (GLCM, LBP)	3,200 images	79.3%

Table 5.4 Comparison of Models for Gender Prediction

STUDY	METHODOLOGY	FEATURES USED	DATASET	ACCURACY
Proposed Work	CNN (22 Layers)	Whorls, Loops, Arches, Ridges, Bifurication	SOCOFing	96.71%
Mensah and Akinyele et al. (2024)	Random Forest	Histogram of Oriented Gradients (HOG)	SOCOFing	90.4%
Das and Dutta (2025)	SVM	Ridge density, minutiae dispersion	3,500 images	88.9%
Shafiq and Hussain et al. (2023)	Logistic Regression	Texture descriptors (GLCM, LBP)	3,200 images	85.7%
Kumar et al. (2025)	Decision Tree	Ridge count ratios, pattern type frequencies	2,800 subjects	78.3%

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

In conclusion, this project has successfully demonstrated the feasibility and effectiveness of using deep learning models for multi-trait human profiling from fingerprint images, enhanced with real-time sensor integration. The development and implementation of a custom Convolutional Neural Network architecture, combined with live fingerprint sensor connectivity, has yielded a robust system capable of accurately predicting blood group, gender, and finger type with remarkable precision in real-time scenarios. The achieved accuracy rates of 97% for blood group classification and 96.74% for combined gender and finger-type classification, coupled with seamless sensor integration, underscore the practical viability of this approach for immediate field applications. The research highlights several significant accomplishments in the domain of biometric analysis and sensor technology integration. The creation of a specialized CNN model, tailored specifically for fingerprint pattern recognition and optimized for real-time sensor data processing, has proven superior to conventional approaches. The successful integration of fingerprint sensors with the deep learning framework enables immediate capture and analysis, eliminating the need for manual image uploads and significantly reducing processing time. This hardware-software synergy represents a crucial advancement in making biometric profiling truly instantaneous and field-deployable.

The system's architecture demonstrates innovative approaches to handling live sensor data streams while maintaining high accuracy standards. The implementation of optimized data pipelines ensures that fingerprint images captured by the sensor are immediately preprocessed, enhanced, and fed into the classification model without noticeable latency. This real-time capability, combined with the non-invasive nature of fingerprint analysis, presents a substantial advantage over conventional blood testing methods, particularly in scenarios requiring rapid biometric profiling such as emergency medical response, law enforcement operations, and high-security access control. The practical applications of this sensor-integrated system are

extensive and impactful. In forensic investigations, the ability to obtain immediate preliminary biometric information at crime scenes can significantly accelerate investigative processes. For emergency medical services, real-time blood group determination without blood sampling can be life-saving in critical situations. Security systems benefit from enhanced authentication that combines traditional fingerprint identification with additional biometric traits, providing multi-layered security verification. The technology effectively establishes that fingerprints, when combined with appropriate sensor technology and advanced machine learning, can serve as a comprehensive source of biometric information that extends far beyond simple identity verification.

6.2 FUTURE SCOPE

While the proposed system performed effectively with scanned fingerprints, a few limitations were observed when using ink-stamped inputs, mainly due to inconsistent print quality and uneven ridge impressions. To overcome these challenges, future work can focus on incorporating advanced sensor technologies such as multi-spectral fingerprint sensors, which can capture both surface and sub-surface ridge details for greater accuracy. The use of portable or wireless fingerprint sensors would also enhance the system's adaptability, allowing deployment in mobile, forensic, and remote environments. Furthermore, integrating sensor fusion techniques that combine fingerprint data with other biometric modalities like palm vein or iris recognition could strengthen reliability and expand the system's applications.

In addition, research into continuous and embedded authentication could enable seamless verification directly through device-embedded sensors, providing real-time and contactless user identification. Future developments may also explore low-power, energy-efficient sensors suitable for field use and blockchain-based data logging to enhance security and traceability. Combining these advancements with edge computing and standardized sensor APIs would support faster processing, improved privacy, and broader compatibility across systems. Such enhancements will advance this work toward a more robust, scalable, and intelligent fingerprint-based biometric platform for practical and high-security applications.

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